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GEOPHYSICS

A New 254 Ma Paleomagnetic Pole of the East European Platform: The Moscow Syneclise, the Sukhoborka and Sosnovka Sections

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Abstract—The results of paleomagnetic and rock magnetic studies of the upper Permian red beds from the Sukhoborka and Sosnovka sections of the south-eastern of the Moscow syneclise on the right bank of the Vetluga River in Nizhny Novgorod Region, Russia, are presented. The arguments in favor of the primary nature of the characteristic magnetization component are indicated. The first late Permian (254 Ma) paleomagnetic pole dated precisely for the sedimentary Permian–Triassic complex in the eastern part of the East European Platform is calculated: 56.731° N, 45.748° E, $N = 44$, plat = 55.3°, plong = 158.1°, dp/dm = 3.5°/5.1°, paleolat = 34.6°.

Keywords: paleomagnetism, East European Platform, Russian Plate, Permian, paleomagnetic pole

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The paleomagnetic poles of the East European Platform (EEP) for the interval of 260–240 Ma that meet the criteria of quality and reliability are not numerous and are obtained primarily based on the sedimentary complexes of "stable" (non-Alpine) Europe [1–3] without accurate geochronological reference. This significantly complicates the development of the Permian–Triassic segment of the apparent polar wander path of the EEP, which is the basis for solving of urgent tasks, in particular, for estimating the scale of post-Paleozoic relative movements of the EEP and Siberia [4, 5], as well as Laurentia (Eurasia). The terrigenous Permian–Triassic complex in the Russian Plate of the EEP have been studied long and actively by the paleomagnetic method, and a series of paleomagnetic poles has been calculated for it in recent years [6–8]; however, their age is determined by the stratigraphic position (which is often ambiguous) of the corresponding sections and cannot be considered to be sufficiently accurate. Therefore, an important task is to obtain new precise paleomagnetic poles for the EEP, the age of which is estimated highly precisely by isotope–geochronological methods.

The study object is the Sukhoborka section of upper Permian sedimentary rocks (Nizhny Novgorod Region, 56.731° N, 45.748° E) located in the southeastern margin of the Moscow syneclise of the EEP, where the ash layer with primarily igneous zircons had been discovered earlier; their age determined by the U–Pb CA-ID-TIMS method was 253.95 ± 0.06 Ma [9]. The section with a total thickness of ~10 m (Fig. 1) is composed of continental, mostly alluvial deposits and is represented by variegated sands with thin layers of red siltstones and clays. The upper portion of the section is characterized by the remains of ostracods, conchostracans, bivalves, fish, and tetrapods from the Nefyodovian regional stage of the Upper Vyatkian [9–11]. The ash layer is located in the middle part of the section and is represented by a layer of maroon (cherry) bentonite clays 11–12 cm thick. A total of 37 oriented samples were selected from the ash layer: 18 samples were placed into quartz vials and 19 samples, into plastic containers. In addition, 36 oriented samples were selected from the rocks that over- and underlie the ash layer; the total sampling interval for the Sukhoborka section was 12 m.

During the field seasons of 2020 and 2022, the lowest part of the Sukhoborka section was below the water level of the Vetluga River; therefore, the corresponding stratigraphic interval, as well as the overlying rocks were sampled in the adjacent Sosnovka section located 3.1–4.0 km downstream along the river (56.725° N,

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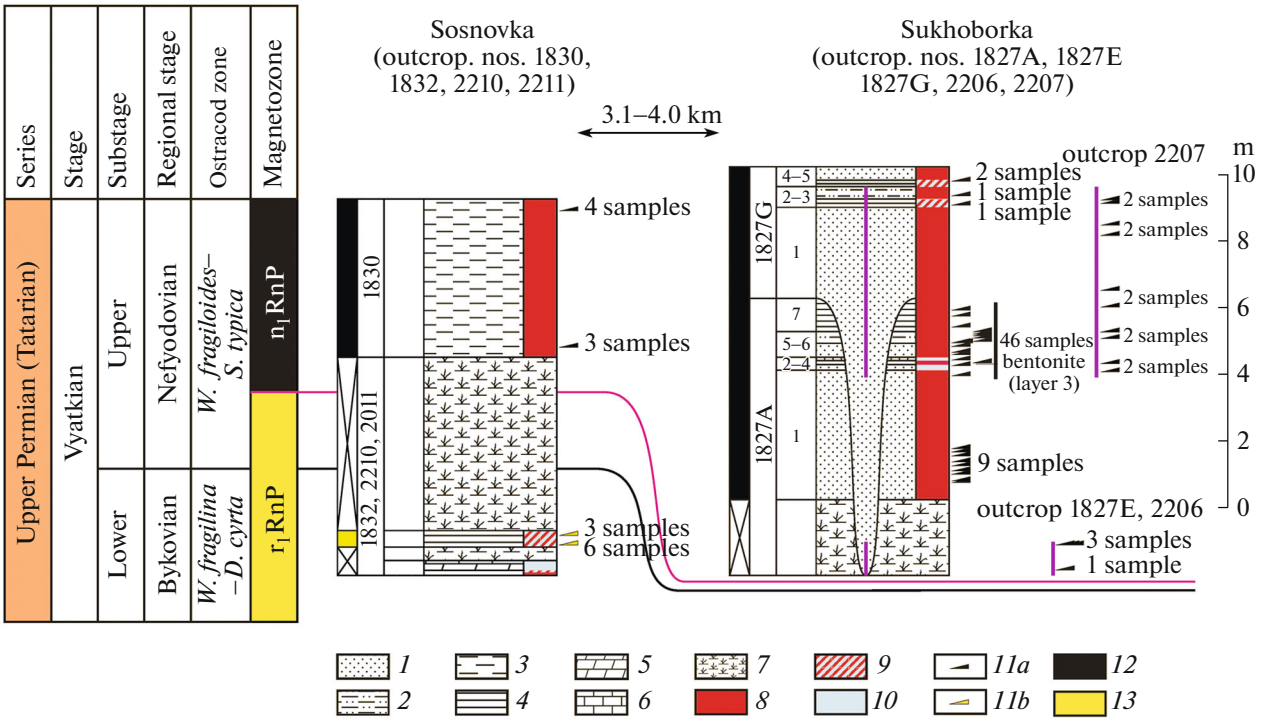


Fig. 1. The intervals of sampling of the Sukhoborka and Sosnovka sections and the number of samples selected. The numbers of outcrops are indicated by V.K. Golubev et al. [10]. Designations: (*1*) sandstone; (*2*) sandy siltstone; (*3*) clayey siltstone; (*4*) clay; (*5*) marl; (*6*) limestone; (*7*) talus; (*8*) red beds; (*9*) variegated beds; (*10*) gray-colored beds; (*11*) levels of collecting paleomagnetic samples with the NRM characteristic component: (*a*) of normal polarity, (*b*) of reversed polarity; (*12*) zones of normal (N) polarity; (*13*) zones of reversed (R) polarity.

45.806° E). The lower part of this section (Fig. 1) is characterized by ostracods in the upper part of the Bykovian regional stage of the Lower Vyatkian [9–11]. In the Sosnovka section, 16 oriented samples were selected from the interval of the section with a total thickness of about 10 m. In total, the paleomagnetic collection consisted of 89 oriented samples.

METHODS AND RESULTS OF STUDIES

The paleomagnetic samples were collected manually with a knife, a shovel, and a geological hammer and were placed into plastic and glass containers; the samples were oriented with a magnetic compass, and the correction for local declination was calculated by the IGRF model (13th generation). The rock magnetic and paleomagnetic laboratory studies were conducted using the equipment of the Shared Research Facilities Center "Petrophysics, geomechanics and paleomagnetism" of Schmidt Institute of Physics of the Earth, RAS [12], according to the modern procedure [13]. For all cube-shaped samples, we measured the anisotropy of magnetic susceptibility, which revealed the planar magnetic fabric typical of sedimentary rocks. The results of the experiments conducted for the most typical samples on estimating the temperature dependence of magnetic susceptibility and saturation magnetization, as well as the parameters of six hysteresis loops and the normal magnetization curves indicate the occurrence of three magnetic minerals in the rocks, such as magnetite, maghemite, and hematite. This conclusion is confirmed by X-ray phase analysis: in the ash layer, the carriers of natural remanent magnetization (NRM) are maghemite (69%) and hematite (31%), while in siltstones and sandstones they are magnetite (74–84%) and hematite (16–26%).

The cube-shaped samples with a 2 cm edge and in the glass vials underwent detailed (up to 14 steps) thermal demagnetization (TD) until complete demagnetization, while the samples in the plastic containers were subject to alternating field (AF) demagnetization with an amplitude up to 200 mT. For the samples from the ash layer, AF treatment was ineffective; in the other cases, AF treatment destroyed up to 90% NRM. The principal component analysis of the results of both types of magnetic treatment makes it possible to identify the most stable component of NRM—the characteristic remanent magnetization (ChRM) that is destroyed in the temperature interval of 300–630°C (Figs. 2a, 2b) or in the interval of the amplitudes of an alternating magnetic field of 20–110 mT (Fig. 2c). Importantly, the polarity of ChRM in the Sukhoborka section is normal (N) (Fig. 2a) and is normal and

Table 1. Paleomagnetic directions and poles derived from the Sukhoborka composite section (Sukhoborka and Sosnovka sections)

Object	Paleomagnetic direction				
	n	D, deg	I, deg	K	α95, deg
Alternating magnetic field treatment (AF)	34	36.7	62.1	39.5	4.0
Thermal demag. (TD), N-polarity	24	36.0	56.5	28.7	5.6
Thermal demag. (TD), R-polarity	20	223.8	−51.2	72.9	3.8
Thermal demag. (TD)	**44**	**39.8**	**54.1**	**37.7**	**3.6**
Thermal demag. (TD) at $f = 0.6$	44	39.8	66.5	37.7	3.6
Thermal demag. (TD) at $f = 0.9$	44	39.8	56.9	37.7	3.6
Sukhoborka paleomagnetic pole:					
slat = 56.731°, slong = 45.748°, plat = 55.3°, plong = 158.0°, dp/dm = 3.5°/5.1°, paleolat = 34.6°					
Sukhoborka paleomagnetic pole at $f = 0.6$:					
plat = 65.2°, plong = 136.8°, dp/dm = 4.9°/5.9°, paleolat = 49.0°					
Sukhoborka paleomagnetic pole at $f = 0.9$:					
plat = 57.5°, plong = 154.8°, dp/dm = 3.8°/5.2°, paleolat = 37.5°					
Sukhoborka-N pole: plat = 59.1°, plong = 159.8°, dp/dm = 5.9°/8.1°, paleolat = 37.1°					
Sukhoborka-R pole: plat = 51.1°, plong = 156.4°, dp/dm = 3.5°/5.1°, paleolat = 31.9°					

n is the number of single paleomagnetic directions (the number of samples); D and I are declination and inclination, respectively; K is concentration parameter; α95 is the radius of the 95% confidence circle; slat and slong are the coordinates of sampling site; plat and plong are the latitude and longitude of the paleomagnetic pole; dp/dm are the semiaxes of the 95% confidence oval; paleolat is the paleolatitude of the sampling site; AF and TD are the alternating magnetic field and thermal demagnetization, respectively; N and R are the normal and reversed polarities of the characteristic remanent magnetization (ChRM); *f* is the inclination shallowing coefficient.

reversed (R) in the Sosnovka section in its upper (Fig. 2c) and lower (Fig. 2b) parts, respectively. Taking into account the biostratigraphic data, normally magnetized sediments correspond to subzone n_1RnP of the regional magnetostratigraphic scale, while reversely magnetized sediments correspond to subzone r_1RnP (Fig. 1).

The distribution of single directions of ChRM isolated in the samples from the Sukhoborka and Sosnovka sections is shown in Figs. 2d and 2e. To calculate the mean paleomagnetic direction for both sections (Fig. 2d; Table 1), we used only the results of thermal magnetic treatment (Fig. 2d), since the ChRM directions obtained from the data of AF-cleaning (Fig. 2e) have higher than expected positive inclinations, which may result from incomplete isolation of the viscous magnetization component. The ChRM directions of normal and reversed polarity are close to antipodality: $\gamma/\gamma_{cr} = 7.0°/7.0°$ [14]. Taking into account the bipolar distribution of the directions of the characteristic magnetization component, the presence of a magnetic fabric typical of sedimentary rocks, and closeness of the calculated mean paleomagnetic direction to the expected one [5–7], we suggest that the nature of ChRM is primary and its age corresponds, to the time of rock accumulation in the Sukhoborka and Sosnovka sections. The thickness of the tested interval of the section infers that the Sukhoborka pole calculated by the mean paleomagnetic direction (Table 1; Fig. 3) is paleomagnetic. Indeed, according to the estimates of sedimentation rates in the sections with similar lithology [15], the sampled 12-m formation accumulated over at least 10 000 yr, and was likely much longer, which allows us to consider the secular variations to be averaged in the paleomagnetic record.

DISCUSSION

The data calculated again raise a question about the reliability and tectonic interpretation of the Permian–Triassic paleomagnetic poles of the EEP obtained using the sedimentary complexes of the Russian Plate [6–8] (Fig. 3). Despite being almost of the same age with a late Permian (254 Ma) Sukhoborka pole, these poles differ from it significantly lying at distances up to 30° of arc of the great circle to the SE (Fig. 3). The possible causes of such distribution of the poles conventionally include (1) the presence of post-Permian rotations of the EEP or its separate tectonic blocks [4, 5], as well as (2) the effect of inclination shallowing in sedimentary rocks. Detailed analysis of the factors listed requires separate research; however, we note that the Sukhoborka pole does not differ from the middle Permian–Triassic pole of the Siberian Traps [16] but differs from the Siberian middle Permian–Triassic pole calculated by using an expanded sample of single poles [5] (Fig. 3). This indicates that the question concerning the scale of relative movements of the East

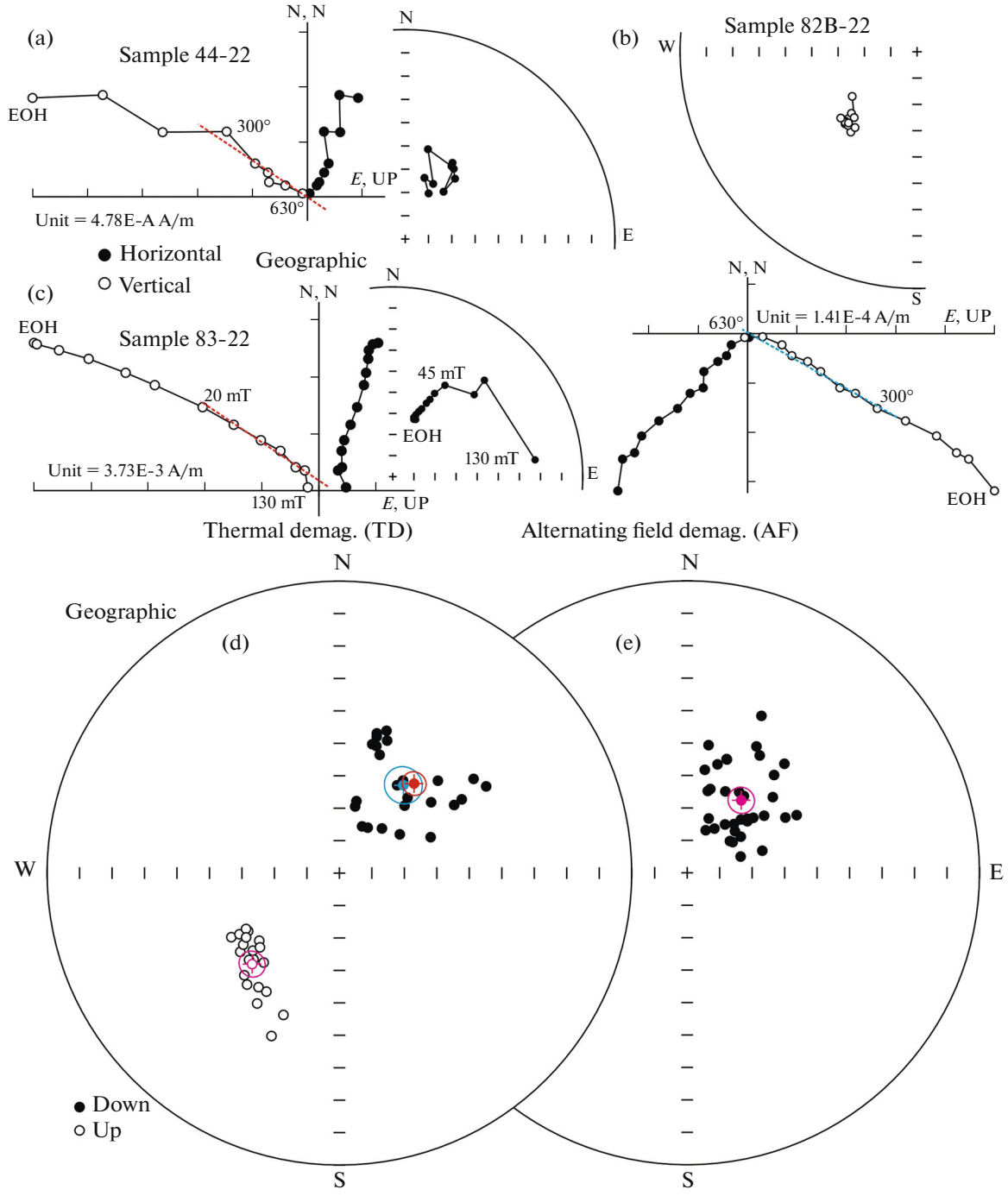


Fig. 2. Results of paleomagnetic studies. (a–c) Zijderveld diagrams and stereograms, illustrating the process of step-wise magnetic demagnetization of the samples: (a, b) thermal demag and (c) alternating field demag. Geographic coordinate system. (a, c) Example of the sample with characteristic remanent magnetization (ChRM) of normal polarity; (b) example of the sample with ChRM of reversed polarity. (d, e) Distribution of single directions of ChRM and mean paleomagnetic directions of the components of normal (blue), reversed (pink, the empty circle), and normal and reversed (red) polarity with 95% confidence circles. Filled (empty) circles in the stereograms are the vector projections onto the lower (upper) hemisphere; filled (empty) circles in the Zijderveld diagrams are the vector projections onto the horizontal (vertical) planes.

European and Siberian platforms during the Mesozoic and the Cenozoic remains open. In addition, the Sukhoborka pole does not differ from the true pole of Europe [1] (Fig. 3), in which the contributions of the effect of inclination shallowing and nondipole components of the main geomagnetic field are excluded. The Sukhoborka pole does not differ either from the poles with ages of 260 and 250 Ma of the apparent

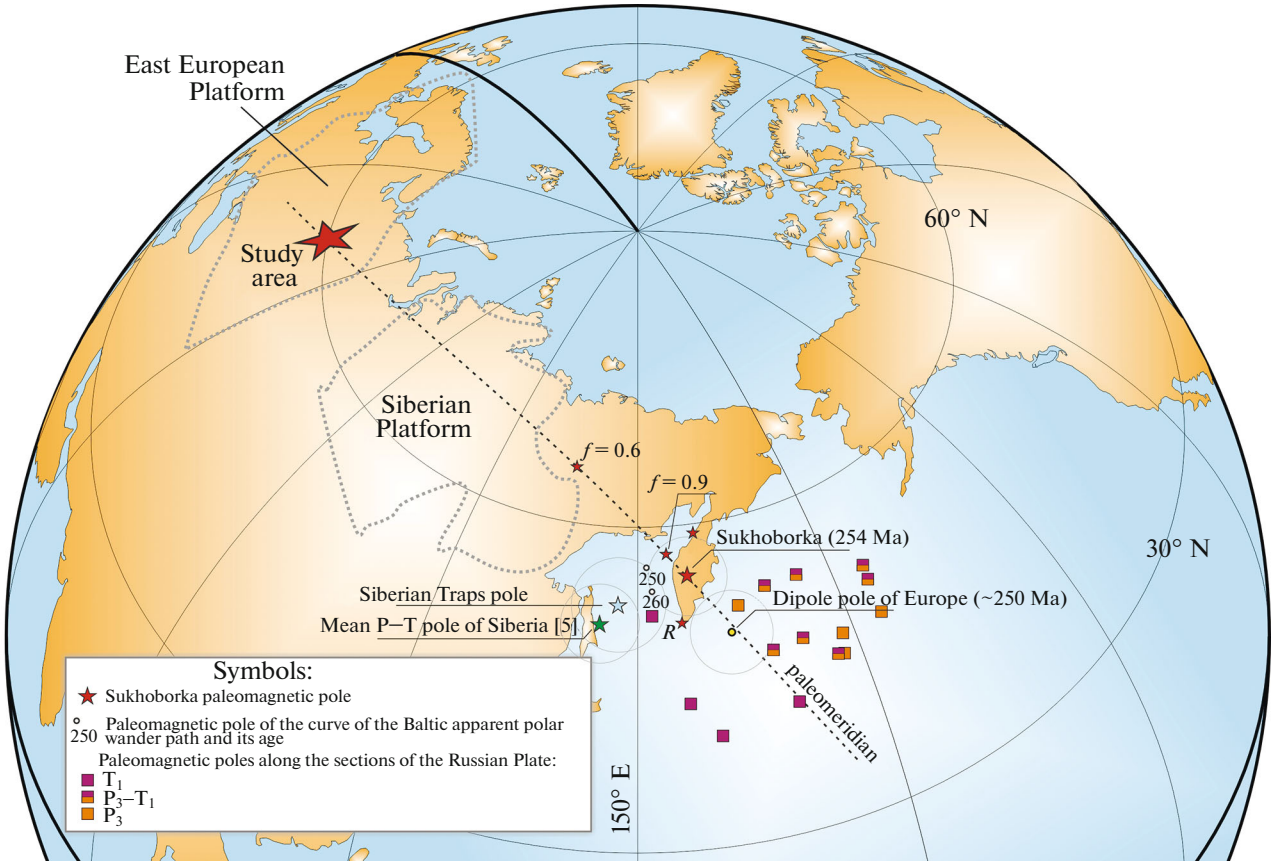


Fig. 3. Sukhoborka paleomagnetic pole of the East European Platform (254 Ma) and the poles discussed in this work: "dipole" pole of Europe [1]; mean P–T pole of Siberia [5]; Siberian Traps [16]; the poles of the apparent polar wander path of the EEP paleomagnetic pole and their ages from [2]; "R" and "N" are the Sukhoborka poles calculated by the directions of reversed and normal polarity, respectively; "*f* = *0.6*" and "*f* = *0.9*" are the Sukhoborka pole corrected for inclination shallowing with the coefficient *f* = 0.6 and *f* = 0.9, respectively. The colored squares show the EEP paleomagnetic poles obtained for the objects within the Russian Plate in [6–8]; the color corresponds to the pole age. The gray circles are the 95% confidence circles.

polar wander path of the Baltic (Fig. 3), upon the calculation of which the paleomagnetic directions derived using the sedimentary rocks were corrected with respect to the effect of inclination shallowing with the coefficient *f* = 0.6 [2]. Concerning the question of inclination shallowing in the sedimentary rocks, we note that the sample size of the single paleomagnetic directions we derived does not allow the use of the statistical *E*/*I* method to estimate the inclination shallowing coefficient in the rocks of the Sukhoborka and Sosnovka sections. However, in case a correction is made for inclination shallowing with the coefficient *f* = 0.6, the Sukhoborka paleomagnetic pole considerably shifts from its expected position (Fig. 3); therefore, the indicated value of *f* cannot be applied to the rocks studied. At the same time, the Sukhoborka paleomagnetic pole allows the use of inclination shallowing coefficient *f* in the rocks of the Permian–Triassic complex from the Russian Plate in the interval of 0.8–0.9 calculated earlier for the sections of Klykovo, Zhukov Ovrag, and Gavrino [6, 7].

We finally note that the authors in [1] consider the incomplete removal of a viscous component during magnetic cleaning to be one of the possible causes for the difference between the Permian–Triassic paleomagnetic poles of the sedimentary complexes in the EEP; therefore, the paleomagnetic directions of reversed polarity, which dominate in the late Permian intervals of the Russian Plate sections, can be shallow. Indeed, as applied to the data obtained, the coordinates of the poles corresponding to the mean directions of ChRM of normal (Sukhoborka-N) and reversed (Sukhoborka-R) polarity of the Sukhoborka–Sosnovka composite section we studied are spaced ~10° apart (Fig. 3). However, such distribution of the poles does not affect the result: the Sukhoborka-N and Sukhoborka-R poles under study differ significantly from most Permian–Triassic poles of the EEP. Nevertheless, the mentioned effect should not be ignored in calculating the paleomagnetic poles obtained by the distributions of the paleomagnetic directions of different polarity.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest.

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